PRASTUTI’26

🥤 AUTOMATED VENDING MACHINE CONTROLLER

Modern campuses rely heavily on automated systems to improve efficiency and reduce human intervention. One such system is the **Automated Vending Machine**, which allows users to purchase products quickly using digital payment or coin-based mechanisms.

Your task is to design a **digital controller for a vending machine** that accepts coins, processes product selection, and dispenses the selected item.

The controller must ensure correct transaction handling, maintain system states, and manage product dispensing using digital logic circuits**.**

System Description

The vending machine offers three products**:**

Product	**Price**
Product A	**₹10**
Product B	**₹15**
Product C	**₹20**

Users insert coins and select a product.
The vending machine must verify whether the **inserted amount is sufficient** and then dispense the selected item.

If the inserted amount is greater than the product price, the system must **return the remaining balance as change**.

Inputs

1. ₹10 Coin Input
2. ₹20 Coin Input
3. **Product Selection Buttons**

- Product A
- Product B
- Product C

4. Reset Signal

Outputs

1. **Product Dispense Signals**
   - Dispense A
   - Dispense B
   - Dispense C
2. Change Return Indicator and amount returned
3. Insufficient Balance Indicator

Tasks

1️⃣ Coin Detection and Encoding

Design a system that detects inserted coins and encodes them into digital signals.

Points: 25

2️⃣ Transaction Controller

Design a **Finite State Machine (FSM)** that manages vending machine operations.

Possible states include:

- Idle
- Coin Insert
- Product Selection
- Product Dispense
- Change Return

Participants must provide:

- State diagram
- State transition table
- Logic implementation

Points: 40

3️⃣ Product Selection Logic

Implement a system that compares the **inserted balance with product price** and determines whether the product can be dispensed.

Points: 20

4️⃣ Output Control System

Generate appropriate output signals for:

- Product dispensing
- Change return
- Insufficient balance indication

Points: 15

Implementation Rules

The circuit must be implemented **only using digital logic components**, such as:

- Logic Gates
- Flip-Flops
- Counters
- Encoders / Decoders
- Multiplexers / Demultiplexers
- Comparators

❌ Microcontrollers are **not allowed**.

The entire system must be designed and simulated using **Proteus Design Suite**.

⭐ Bonus Challenge (Extra 20 Points)

Verilog HDL Implementation

Participants may additionally implement the **Vending Machine Controller using Verilog HDL**.

The Verilog design should include:

- FSM-based transaction control
- Coin input detection
- Balance counter
- Product selection logic
- Product dispense and change return logic

Participants must submit:

1️⃣ Verilog module(s) implementing the system
2️⃣ A **testbench** verifying system functionality
3️⃣ Simulation waveform results

Bonus Evaluation Criteria

Component	**Points**
Correct Verilog implementation	10
Testbench & simulation results	5
Code structure and modularity	5

Total Bonus Points: **20**

Good luck and happy designing!